Employee Attrition & Performance Analytics

GlobalTech Solutions | HR Analytics | Classification | People Analytics

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1. Executive Summary

GlobalTech Solutions is experiencing a 30% employee attrition rate — double the industry average of 15% — resulting in significant talent loss and rising hiring costs. This project builds a classification model using Logistic Regression and Random Forest to predict which employees are likely to leave. 245 high-risk employees were identified and a priority HR action list was generated. The model achieved a ROC-AUC of 0.722.

2. Problem Definition

Business Problem

- 30% annual attrition rate — double the industry average of ~15%
- Loss of skilled employees causing productivity decline across teams
- Rising recruitment and training costs due to frequent backfills
- No early warning system to identify at-risk employees before they resign
- HR team relies on exit interviews — reactive, not proactive

Key Objectives

- Build a model to predict which employees are likely to leave
- Identify the top factors driving attrition across the organisation
- Analyse the relationship between performance, satisfaction, and attrition
- Segment all 1,000 employees into High / Medium / Low attrition risk
- Provide HR with a prioritised action list for immediate intervention
- Recommend targeted retention strategies per risk segment

3. Dataset Description

Dataset Attribute Details

Total Employees 1,000

Total Features 18 (16 input + 1 ID + 1 target)

Key Features Monthly Income, Overtime Hours, Job Satisfaction, Years at Company, Absenteeism

Target Variable Attrition (No = Stayed, Yes = Left)

Class Distribution 700 Stayed (70%) | 300 Left (30%)

Missing Values None — dataset fully clean

Departments IT, Sales, HR, Finance, Operations, Support

4. Methodology & Data Preprocessing

Data Preprocessing

• Label Encoding applied to all text columns: Gender, Department, Role, Education, Attrition

• Dropped EmployeeID — not a predictive feature

• Train/Test split: 80/20 with stratified sampling to maintain 30% attrition ratio in both sets

• StandardScaler applied to normalise all numeric features

Modelling Approach

• Model 1 — Logistic Regression: interpretable baseline with probability output

• Model 2 — Random Forest (100 trees): stronger ensemble model

• Evaluation: Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix

• Feature importance from Random Forest used to rank attrition drivers

• All 1,000 employees scored with attrition probability (0–100%)

• Three-tier risk segmentation: Low (<35%), Medium (35–60%), High (>60%)

5. Results & Model Performance

KPI / Metric Target / Benchmark

Logistic Regression — Accuracy 74.5%

Logistic Regression — ROC-AUC 0.714

Random Forest — Accuracy 75.0%

Random Forest — ROC-AUC 0.722

Top 3 Attrition Drivers Monthly Income, Overtime Hours, Years at

Company

Highest Risk Department Sales (35.9% attrition rate)

High Risk Employees Flagged 245 out of 1,000 (24.5%)

Risk Segmentation

Risk Tier Action

High Risk (>60% prob) 245 employees — immediate HR intervention

required

Medium Risk (35–60%) 74 employees — monitor and engage

proactively

Low Risk (<35%) 681 employees — standard engagement

programme

6. Business Recommendations

• Conduct retention interviews with all 245 HIGH RISK employees this quarter

• Introduce overtime cap at 20 hours/month — overtime is the 2nd strongest attrition driver

• Expand remote work flexibility — linked to higher job satisfaction scores

• Create structured promotion track — PromotionLast5Years scored very low across leavers

• Increase training budget — low training hours strongly correlated with exit

• Focus immediate HR resources on the Sales department (highest attrition at 35.9%)

• Introduce stay bonuses for employees at the 2–3 year tenure mark (highest risk window)

• Run this model monthly on fresh HR data — alert when any employee crosses 60% threshold

7. Conclusion & Future Work

• Random Forest (ROC-AUC 0.722) outperforms Logistic Regression for attrition prediction

• Monthly Income, Overtime, and Tenure are the three dominant attrition drivers

• 245 employees are currently at high risk — immediate intervention can prevent costly exits

• Future work: XGBoost model, real-time HR dashboard, NLP on exit interview feedback

• Target: reduce attrition from 30% to below 20% within 12 months of intervention